



Term Evaluation (Odd) Semester Examination September 2025

Roll no.....

Name of the Course: MCA AIDS

Semester: I

Name of the Paper: Advance Database Management System

Paper Code: TMD 104

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Data Independence is achieved in modern databases using schemas, Exemplify the schema structure to achieve the above? **CO1**

OR

b. If you are a database analyst for Oracle Corporation, as a stake holder of the corporation to train the customer on database environment, list the functionalities and modules in DB. **CO1**

Q2. (10 Marks)

a. i. Define strong and weak entities with suitable examples. Analyze why weak entities require the existence of a strong entity for identification. **CO1**

ii. Explain the concept of Union compatibility with respect to set operations in relational algebra. Give example to justify your answer

OR

b. Why keys are important in relational model? Explain about super key, candidate key, primary key and alternate key with proper example. What are integrity constraints in a relational model? Describe domain, entity integrity, and referential integrity constraints and explain their importance in maintaining database consistency. **CO1**

Q3. (10 Marks)

a. Explain the concepts of specialization and generalization in the Enhanced Entity Relationship (EER) model. How do they differ? Also define the various constraints on specialization and generalization. Give examples of each type. **CO1**

OR

b. Write queries for the following in relational algebra using the database below:

STUDENT (SID, Name, DeptID, Age)

DEPARTMENT (DeptID, DeptName)

- (i) List the Name and Age of students who are older than 21 years
- (ii) Display the SID, Name, and DeptID of students belonging to department 'CSE'.
- (iii) Show the Name of all students along with their Department Name.
- (iv) List name and age for all students whose SID lies in the range from 20 to 30.
- (v) List the department name for deptid 40



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Q4. (10 Marks)

a. Explain the different steps briefly that are involved in processing a high-level query. CO2

OR

b. What is normalization? Why it is required. Define 1NF, 2NF and 3NF with example. CO2

Q5. (10 Marks)

a. Define decomposition in relational database design. Differentiate between lossy and lossless decomposition with the help of suitable examples. Why is lossless decomposition preferred? CO2

OR

b. Suppose you are given the following requirements for a simple database for the National Hockey League (NHL): Col (20 Marks)

- The NHL has many teams,
- each team has a name, a city, a coach, a captain, and a set of players,
- each player belongs to only one team,
- each player has a name, a position (such as left wing or goalie), a skill level, and a set of injury records,
- a team captain is also a player,
- a game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 2017) and a score (such as 4 to 2).

Construct a clean and concise ER diagram for the NHL database. List your assumptions and clearly indicate the cardinality mappings as well as any role indicators in your ER diagram. Minimize the number of tables and map the ER diagram into corresponding tables.